Syllabus
Second Year
B.Tech in Electronics and Telecommunication Engineering
(Programme commenced from AY 2023-24)
(Department of Electronics and Telecommunication Engineering)

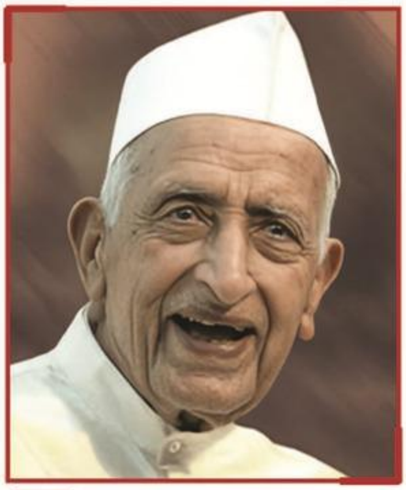
From
Academic Year 2024-25
(SVU-KJSCE 2.0)

(Approved by BOS dated 30th April 2024, FOET dated 2nd July 2024 and AC dated 4th July 2024, Item No. 12.04)

SOMAIYA VIDYAVIHAR UNIVERSITY

Somaiya TRUST

Our Vision



Padma Bhushan
Shri Karamshi Jethabhai Somaiya
Our Founder

Padma Bhushan Shri Karamshi Jethabhai Somaiya's first initiative in education was the founding of a school in 1942 in rural Maharashtra, to provide quality holistic education. It was founded on the belief that education is an important pillar of nation-building with the power to change lives and that it is the duty of the privileged to provide education to the needy.

He later founded the Girivanvasi Pragati Mandal, The K J Somaiya Medical Trust, the Girivanvasi Education Trust and sister institutions to make great citizens of India and the world. In the words of Swami Vivekananda, "We want that education by which character is formed, strength of mind is increased, the intellect expanded, and by which one can stand on one's own feet." We have now grown into a multi-disciplinary and multi-campus education institution with over 1500 faculty, and 39000+ students.

ज्ञानादेव तु कैवल्यम्।

KNOWLEDGE ALONE LIBERATES

Our motto is: ज्ञानादेव तु कैवल्यम्। Knowledge alone liberates. Liberates from poverty, from hunger. Also to liberate one from the attachments that bind us to small-mindedness. Knowledge also provides opportunity. To make the life lived more meaningful. In the service of one's family, one's community, one's समाज, country, and indeed the world. Bearing in mind that there is no religion other than the life lived in the service of humanity.

न मानुषात् परो धर्मः।

We will strive to provide access and opportunity to build a more inclusive society.

Our education in any subject will reflect its timeless fundamentals, its current context, and its applications. There is so much scientific discovery taking place, at the intersection of fields, in biology, computing, medicine, the social sciences and everywhere else. We will provide students and faculty with an environment to engage with this world, discover new truths, and make new applications to create and share knowledge.

Our education will also be experiential. With projects that are 'real' and those that complement the learning inside the classroom. Our students and faculty will be at the cutting edge of change, to incubate companies, to create NGOs, and pursue any field of their passion.

Our education will also be holistic. Sports and physical exercise must be a firm part of the curriculum. For students to develop a love for sports, for recreation, for health, for teamwork, and for competition. Our education will also instil an appreciation for art, culture, nature and biodiversity.

अभ्यासेन तु कौन्तेय वैराग्येण च गृह्यते।

In the Bhagavad Gita, Arjun asks Krishna how is one to control one's mind, which is as fleeting as the wind. Krishna responds that it can only be done through practice and discipline. अभ्यासेन तु कौन्तेय वैराग्येण च गृह्यते। We will strive to teach our students to learn to stay calm in our turbulent world. And our education will also include the ancient Indian tradition, its culture, its depth, and its knowledge. We must keep the connection with our mother tongue and our languages. Languages are storehouses of culture, and the loss of a language takes with it much learning, stored through it over the ages.

Finally, our education will help students lead a full life, and to fall in love with life. Our dream is to build a world-class research and teaching institution, that is global in the reach of its ideas, and universal in its service. Welcome to our community.

Our Mission

To nurture excellence and provide freedom of possibilities in education and research to foster a culture of creativity, innovation, leadership, responsible citizenship, service, and all-round growth.

Premble

From the Desk of Dean Faculty of Engineering and Technology:

In the era of technological revolution, engineering education must evolve to keep pace with the dynamic demands of industry and society. Our engineering institute is committed to fostering a learning environment that nurtures innovation, creativity, and a profound understanding of engineering principles. The **National Educational Policy 2020 (NEP 2020)** framed by the Government of India recommends a holistic, inclusive, and flexible approach to ensure equitable access to quality education across all levels, promote multidisciplinary research, and impart skill-based education with integration of technology.

Somaiya Vidyavihar, with its esteemed legacy in education, has consistently upheld the values of excellence, inclusivity, and innovation. Applicable for **Somaiya Vidyavihar University (SVU)'s** undergraduate engineering programs, the **SVU Scheme 2023** presented here is aligned with the transformative vision of Somaiya Vidyavihar as well as NEP 2020 to cultivate a holistic, experiential, and interdisciplinary approach to engineering education. The **salient features** of the scheme include:

Professional Core and Elective Courses: The curriculum includes state-of-the-art courses that cover both the fundamentals and emerging trends in respective branches of engineering. With an optimal balance between theoretical knowledge and practical application, core courses provide a strong foundation in essential engineering principles, while elective courses offer flexibility for students to explore and specialize in areas of interest.

Open Elective Courses: Recognizing the importance of interdisciplinary knowledge, the curriculum includes a diverse range of Open Electives categorized into four types: Open Elective Technology (OET), Open Elective Humanities, Open Elective Management (OEHM), and Open Elective Generic (OEG). These courses, offered at institute-level, enable students to expand their knowledge across various disciplines, fostering a versatile skill set and adaptability in an ever-evolving global landscape.

Innovation and Project-based Learning (PBL): The curriculum engages students in innovation and PBL through ideation, mini and major projects right from the first year to the final year of engineering. With diverse projects, collaboration, and field work/community engagement initiatives, students gain a profound understanding of engineering concepts and contribute through innovative solutions to the Sustainable Development Goals (SDGs), societal challenges and advancements.

Learning-by-Doing: The curriculum places emphasis on exposure courses through Skill-Based Learning (SBL) and Activity-Based Learning (ABL), focusing on responsibilities towards society, problem-solving abilities, leadership and teamwork, motivation for life-long learning, etc.

Elements of the Indian Knowledge System: The curriculum incorporates aspects of the Indian Knowledge System that emphasize on drawing insights from ancient wisdom and rich intellectual heritage of India to address modern challenges.

Internships and Research: Enabling students to gain industry insights and enhance their employability, the curriculum integrates flexible internship opportunities in Semester VII or VIII, allowing students to gain hands-on experience in industries, government sectors, NGOs, and MSMEs. Alternatively, they can opt for a specialized research project and courses in Semester VIII. Besides this Semester-long Internship, all the students are required to complete a mandatory 10-week internship over four years, with a maximum of 4 weeks dedicated to socially relevant internships and a minimum of 6 weeks in technical domains.

Learning through MOOCs: The curriculum leverages and promotes Massive Open Online Courses (MOOCs) to offer students flexible and diverse learning opportunities. Complementing on-campus education, students can learn through MOOCs for Open Electives – OET and OEHM during the Pre-final and Final Year, as well as Professional Core courses during their Internship.

Student Exchange Programs: The curriculum also offers student exchange programs that promote global exposure and cross-cultural learning, elevating academic and personal growth. Interested students can participate in the Student Exchange Programs as an alternative to the semester-long internship. Credits from the foreign university where they study will be transferred, providing them with an opportunity to experience different educational systems, cultures, and perspectives.

Minors Courses: Students can expand their academic horizons by pursuing minors in disciplines other than their major, earning additional credits. These minor courses provide an opportunity to acquire multidisciplinary knowledge, significantly enhancing their versatility and adaptability in the professional world.

Honors Courses: For high-achieving students, the SVU 2023 scheme offers Honors courses that delve deeper into specialized topics and gain additional credits for the same. These advanced courses align with high-end industry standards and provide an enriched learning experience, offering multiple opportunities to expand knowledge and expertise in areas of interest.

This forward-thinking SVU 2023 scheme is designed to equip our graduating engineers to emerge as innovative leaders, capable of addressing global challenges and contributing to the advancement of society. Our Boards of Studies, comprising experts in different disciplines, have meticulously designed syllabus for various programs under this SVU 2023 Scheme. We are confident that the joint efforts of the faculty, alumni, students, industry experts, and all the stakeholders will strengthen the academic, research, and entrepreneurial culture of our institution, reinforcing K. J. Somaiya College of Engineering's position as one of the premier engineering institutions in the nation and a top choice for engineering aspirants.

Dr. S. K. Ukarande

Dean – Faculty of Engineering and Technology
Somaiya Vidyavihar University, Mumbai

Preface

The department of Electronics and Telecommunication (EXTC) was established in 1996. Since its commencement, the primary objective of the department has been to impart quality education, training and promoting research at both undergraduate and postgraduate level. The department focuses on various areas of Electronics and Telecommunication Engineering with broad emphasis on design aspects of electronic and communication systems. The National Board of Accreditation (NBA) has accredited the department thrice from 2005-2008, 2009-2012 and 2013-2015. The college also received the National Assessment and Accreditation Council (NAAC) accreditation in May 2017 with grade “A” in tier I.

EXTC department has all the laboratories with modern equipment. The department has also taken initiative to establish specialized laboratories with industry support in multidisciplinary domains such as networking, RF and Internet of Things. Currently the department has Cisco Centre of Excellence in networking and Texas Innovation Centre under which, various academic & research activities are conducted. The department has always been on a high growth path and has experienced and dedicated faculty members with a strong commitment to engineering education. The major areas of faculty expertise include Basic Electronics, Communication Systems, Computer Networks, Control Systems, Digital Signal Processing, Image Processing, Computer Vision, Microprocessors, Embedded Systems, Instrumentation, RF & Microwaves and VLSI Systems.

More than 450 students are seeking professional education in the department under the supervision and guidance of all faculty and staff members. The UGC conferred autonomy status to college in 2013 and subsequently the college introduced its first autonomous syllabus, KJSCE 2014 from academic year 2014. Furthermore, in the pursuit of imparting quality and holistic education, K. J. Somaiya College of Engineering became a constituent college of Somaiya Vidyavihar University (2019). Accordingly, the department enhanced its curriculum so as to match the requirements of the contemporary world.

The revised curriculum (SVU 2023) is designed with an objective of preparing students to meet new industry challenges and make them aware of the recent advances in technological field. This is achieved by emphasizing on:

- Strong technical training mechanisms
- Skill enhancement in design, development, analysis and testing of microelectronic circuits and communication systems
- Latest trends in simulation and their practices
- Promoting Interdisciplinary / Multidisciplinary activities through courses and project based learning
- Contemporary human resource management and development

The curriculum SVU 2023 is apt for students who wish to become professional engineers in the field of Electronics and Telecommunication Engineering thereby finding solutions to practical problems in real world. All theory courses are supported by tutorials / laboratory and special laboratory courses for developing programming skills. Special attention has been given on coverage of fundamental professional core courses from all thrust areas of department. The same are equally supported by a strong laboratory component and considerable project work.

SY syllabus in particular covers the basic courses as required in electronics and communication field. It also creates a foundation for interdisciplinary activities as required in today’s world. Moreover special attention has been given to utilization of workshop and laboratories by students for the extensive project work which is an additional feature of the program. Thus, with proper syllabus formulation, we intend to achieve the program objectives, ultimately leading to the attainment of the mission and vision of the department.

HOD EXTC

Board of Studies in Electronics and Telecommunication Engineering

Dr. Bharati. A. Singh	:	Chairman
Dr. Kumar Appaiah	:	Academician Member
Mr. Shashank Karmarkar	:	Industry Member
Dr. Padmaja Joshi	:	Research Institute Member
Mr. Prateek Mehta	:	Alumni Industry Member
Dr. Ramesh Karandikar	:	Faculty Member
Dr. Maruti Zalte	:	Faculty Member
Dr. Kiran Ajetrao	:	Faculty Member

Dr. S. K. Ukarande : Dean – Faculty of Engineering and Technology

Vision of the Department:

To become centre of excellence for creating competent engineers with evolving technical skills, leadership qualities with human values to pursue excellence in professional field

Mission of the Department:

Providing quality education to

- Develop technical skills, soft skills and professional ethics to cater the needs of industry.
- Promote research and creativity in Engineering and Technology.
- Inculcate awareness towards societal needs and environmental issues**.**

Program Educational Objectives (PEO) of the Department:

A graduate of Electronics and Telecommunication Engineering will

PEO1. Excel in professional career by adapting emerging technologies.

PEO2. Pursue higher education, research, address environmental issues with ethical practices.

PEO3. Solve real life problems in a team as a member or as a leader.

Program Outcomes (PO) – Common to all disciplines

PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, cultural, environmental, health, safety and legal issues relevant to the professional engineering practice; understanding the need of sustainable development

PO7. Multidisciplinary Competence: Recognize/ study/ analyze/ provide solutions to real-life problems of multidisciplinary nature from diverse fields

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSO)

PSO1: Pursue higher studies in the field of Communication, Embedded systems and allied domains

PSO2: Acquire, demonstrate industry relevant competencies from core telecommunication discipline and emerging fields

Acronyms used:

1. Acronyms for category of courses and syllabus template

Acronym	Description	Acronym	Description
BS	Basic Science Courses	**CA**	Continuous Assessment (Theory Course)
ES	Engineering Science	**ESE**	End Semester Exam
HS	Humanities, Social Sciences and Management Courses	**ISE**	In- Semester Examination
PC	Professional Core Courses	**IA**	Internal Assessment
PE	Professional Elective courses	**LAB/TUT CA**	Continuous Assessment of Laboratory/Tutorial
OET	Open Elective – Technical	**TH**	Theory
OEHM	Open Elective – Humanities and Management	**TUT**	Tutorial
OEG	Open Elective Generic	**CO**	Course Outcome
LC	Laboratory Courses	**PO**	Program Outcome
PR	Project	**PSO**	Program specific Outcome
EX	Exposure Course	**IKS**	Indian Knowledge System

2. Type of Course

Acronym	Description
C	Core Course
E	Elective Course
O	Open Elective Technical
H	Open Elective - Humanities/ Management/
R	Open Elective Generic
P	Project
L	Laboratory Course
T	Tutorial
X	Exposure course
W	Workshop
I	Indian Knowledge System

3. Eight Digit Course code e.g. 216U03C301

Acronym Serially as per code	Description
2	SVU-2023 Second Revision
16	College code
U	Alphabet code for type of program
03/06	Program code/ Common to all
C	Type of course
3	Semester number (Semester III)
01	Course serial number

SEMESTER III (With Effect from 2024-2025)

Teaching and Credit Scheme

Course Code	Course Category	Name of the Course	Teaching Scheme TH-PR-TUT	Total (hrs.)	Credit Scheme TH-PR-TUT	Total Credits
216U03C301	BS	Mathematics for Communication Engineering	3 – 0 – 1	04	3 – 0 – 1	04
216U03C302	PC	Basic Electronic Circuits	3 – 0 – 0	03	3 – 0 – 0	03
216U03C303	PC	Digital Logic Design	3 – 0 – 0	03	3 – 0 – 0	03
216U03C304	PC	Microprocessor and Microcontroller	3 – 0 – 0	03	3 – 0 – 0	03
216U03C305	PC	Electrical Networks	3 – 0 – 1	04	3 – 0 – 1	04
216U06I306	IKS	Indian Knowledge System	2 – 0– 0	02	2 – 0– 0	02
216U03L301	PC	Data Structures and Analysis of Algorithms Laboratory	1– 2 – 0	03	0 – 2– 0	02
216U03L302	PC	Basic Electronic Circuits Laboratory	0 – 2 – 0	02	0– 1 – 0	01
216U03L303	PC	Digital Logic Design Laboratory	0 – 2 – 0	02	0– 1 – 0	01
216U03L304	PC	Microprocessor and Microcontroller laboratory	0 –2 – 0	02	0– 1 – 0	01
		Total	**18– 8 – 2**	**28**	**17 – 5 – 2**	**24**

Evaluation Scheme

Course Code	Course Category	Name of the Course	LAB/ TUT CA	CA		ESE	Total
				IA	ISE		
216U03C301	BS	Mathematics for Communication Engineering	25	20	30	50	125
216U03C302	PC	Basic Electronic Circuits	--	20	30	50	100
216U03C303	PC	Digital Logic Design	--	20	30	50	100
216U03C304	PC	Microprocessor and Microcontroller	--	20	30	50	100
216U03C305	PC	Electrical Networks	25	20	30	50	125
216U06I306	IKS	Indian Knowledge System	--	50		--	50
216U03L301	PC	Data Structures and Analysis of Algorithms Laboratory	75	--	--	--	75
216U03L302	PC	Basic Electronic Circuits Laboratory	50	--	--	--	50
216U03L303	PC	Digital Logic Design Laboratory	50	--	--	--	50
216U03L304	PC	Microprocessor and Microcontroller laboratory	50	--	--	--	50
		Total	**275**	**150**	**150**	**300**	**825**

SEMESTER IV (With Effect from 2024-2025)

Teaching and Credit Scheme

Course Code	Course Category	Name of the Course	Teaching Scheme TH-PR-TUT	Total (hrs.)	Credit Scheme TH-PR-TUT	Total Credits
216U03C401	BS	Probability, Statistics and Optimization Techniques	3– 0 – 0	03	3 – 0 –0	03
216U03C402	PC	Analog Electronics	3 – 0 – 0	03	3 – 0 – 0	03
216U03C403	PC	Communication Systems	3 – 0 – 0	03	3 – 0 – 0	03
216U03C404	PC	Signals and Systems	3– 0 – 1	04	3 – 0 – 1	04
216U03C405	PC	Electromagnetic Field Theory	3 – 0 – 1	04	3 – 0 – 1	04
216U06R406	OE	Open Elective Generic	3 – 0 – 0	03	3 – 0 – 0	03
216U03L401	PC	Probability, Statistics and Optimization Techniques Laboratory	0 –2 – 0	02	0 – 1 – 0	01
216U03L402	PC	Analog Electronics Laboratory	0 –2 – 0	02	0 – 1 – 0	01
216U03L403	PC	Communication Systems Laboratory	0 –2 – 0	02	0 – 1 – 0	01
216U03L404	PC	Advanced Microcontroller Laboratory	0 –2 – 0	02	0 – 1 – 0	01
		Total	**18 – 8 –2**	**28**	**18 –4– 2**	**24**

Evaluation Scheme

Course Code	Course Category	Name of the Course	LAB/ TUT CA	CA		ESE	Total
				IA	ISE		
216U03C401	BS	Probability, Statistics and Optimization Techniques	--	20	30	50	100
216U03C402	PC	Analog Electronics	--	20	30	50	100
216U03C403	PC	Communication Systems	--	20	30	50	100
216U03C404	PC	Signals and Systems	25	20	30	50	125
216U03C405	PC	Electromagnetic Field Theory	25	20	30	50	125
216U06R406	OE	Open Elective Generic		100	-	-	100
216U03L401	PC	Probability, Statistics and Optimization Techniques Laboratory	50	--	--	--	50
216U03L402	PC	Analog Electronics Laboratory	50	--	--	--	50
216U03L403	PC	Communication Systems Laboratory	50	--	--	--	50
216U03L404	PC	Advanced Microcontroller Laboratory	50	--	--	--	50
		Total	**250**	**200**	**150**	**250**	**850**

Note: For detailed syllabus of Open Elective Generic, students need to refer to separate document

List of Open Electives Generic (OEG) as offered for AY 2024-25

Course Code	Course Name
216U6R401	Ancient Indian Iconography
216U6R402	French Culture and Civilization
216U6R403	German Culture and Society
216U6R404	Indian Cinema History and Appreciation
216U6R405	Indian Civilization
216U6R406	Literature of Resistance
216U6R407	Marathi Kavita
216U6R408	Maritime Seafaring and Shipbuilding
216U6R409	Mountains, People and Cultures
216U6R410	Philosophy of Science
216U6R411	The Non-Human Species Depicted in Literature
216U6R412	World Civilizations
216U6R413	Physics and Philosophy

Note:
As per college internship policy, It is mandatory for every student to complete 10 weeks of Internship spanning over the four years of B.Tech Programme over and above the academic credits.
Students can take up internships in community services / socially relevant projects (optional and limited to 4 weeks) and in the technical domain (minimum 6 weeks or more).
Students will be awarded an internship completion certificate along with their graduation.

Course-wise Detailed Syllabus

SEMESTER III

Course Code	Name of the Course			
216U03C301	Mathematics For Communication Engineering			
Teaching Scheme	TH	P	TUT	Total
	03	0	01	04
Credits Assigned	03	0	01	04

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	25	20	30	50	125

Course pre-requisites:

- Applied Mathematics-I, Applied Mathematics-II

Course Objectives:

The objective of this course is to impart knowledge of Transform & Vector Calculus, concepts of Fourier Series and Bessel functions and thereby aid the students to apply the fundamental of mathematics necessary to formulate, solve and analyze Electronics and telecommunication engineering problems in the areas related to Electrical Network analysis, Communication Systems, Control systems, Signals and Systems & Electromagnetic Field Theory.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Find Laplace Transform, Inverse Laplace Transform of function & Apply Laplace Transform to solve Differential Equations.
CO2. Find Fourier Series representation for a periodic function.
CO3. Solve problems on recurrence relation of Bessel function.
CO4. Solve examples using operators grad, div & curl.
CO5. Solve examples to evaluate integrals using Green's Theorem, Stokes' Theorem & Divergence theorem.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	**Laplace Transform**		12	CO1
	1.1	Definition of Laplace Transform, Laplace Transform of sin(at), cos(at), sinh(at), cosh(at), erf(t), Heavi-side unit step, dirac-delta function, Laplace Transform of periodic function		
	1.2	Properties of Laplace Transform @ Linearity, first shifting theorem, second shifting theorem, multiplication by t, division by t , Laplace Transform of derivatives and integrals, change of scale.		
	1.3	Inverse Laplace Transform: Partial fraction method, convolution theorem @		
	1.4	Applications of Laplace Transform: Solution of ordinary differential equations with constant coefficients.		
2	**Fourier Series**		10	CO2
	2.1	Introduction: Definition, Dirichlet's conditions, Euler's formulae		
	2.2	Fourier Series of Functions: Exponential, trigonometric functions, even and odd functions, half range sine and Cosine series		
	2.3	Complex form of Fourier series		
3	**Bessel Functions**		06	CO3
	3.1	Bessel's Differential Equation and Bessel function		
	3.2	Recurrence relation, Properties of Bessel function @.		
4	**Vector Differentiation**		08	CO4
	4.1	Scalar and vector product of three and four vectors and their properties.		
	4.2	Gradient of scalar point function, divergence and curl of vector point function.		
	4.3	Solenoidal and irrotational vector fields.		
#Self-Learning: Co-ordinate Systems and its conversion.				
5	**Vector Integration**		09	CO5
	5.1	Vector Integral: Line integral, Properties of line integral, Surface integral, Volume integrals.		
	5.2	Green's theorem in a plane @		
	5.3	Gauss divergence theorem, Stokes theorem @		
		Total	**45**	

@ Proofs of properties/ theorems are not expected

Students should prepare self-learning topics on their own. Self – learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in IA.

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/ Year
1	B. S. Grewal	*Higher Engineering Mathematics*	Khanna Publications, India	43rd Edition, 2014
2	Erwin Kreyszig	*Advanced Engineering Mathematics*	Wiley Eastern Limited, India	10th Edition 2015
3	P.N.Wartikar & J. N. Wartikar,	*A Text Book of Applied Mathematics, Vol I & Vol II*	Vidyarthi Griha Prakashan,	6th Edition 2012
4	B.V.Ramana	*Higher Engineering Mathematics*	Tata McGrawHill Publishing Company Limited, India	1st Edition 2007
5	N.P.Bali, Dr.Manish Goyal.	*A Textbook of Engineering Mathematics.*	Laxmi Publication, India	9th Edition 2016

Tutorial: Tutorial will consist of at least 10 Tutorials covering entire syllabus out of which 3 to 4 tutorials can be Telecommunication application based. Use of some Programming Language/software is also recommended to conduct Tutorials. Students will be graded based on continuous assessment of their term work

Course Code	Name of the Course					
216U03C302	Basic Electronic Circuits					

Teaching Scheme	TH	P	TUT	Total
	03	-	-	03
Credits Assigned	03	-	-	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	-	20	30	50	100

Course pre-requisites:

- Elements of Electrical and Electronics Engineering

Course Objectives:

The objective of the course is to impart fundamental knowledge and applications of semiconductor devices like P-N junction diode, BJT and FET. This course aims to build a foundation for DC analysis, biasing circuits of BJT, FET and small signal analysis of mid frequency range amplifiers using hybrid π model.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Analyze and design Diode circuits.
CO2. Analyze the transistor circuits for DC operation.
CO3. Understand working of transistor as an amplifier.
CO4. Analyze transistor amplifier for small signal operation for mid frequency range.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	**Diode Applications**		07	CO1
	1.1	Review of p-n junction diode characteristic and current-voltage relationship		
	1.2	Voltage transfer characteristics, series and shunt clipper, diode series and shunt clamper		
	1.3	Working and applications of Varactor diode, Schottky diodes		
2	**Bipolar Junction Transistor (BJT)**		09	CO2
	2.1	Review of BJT characteristics, regions of operation. DC load line, Early effect and thermal runaway		
	2.2	DC Analysis and design of Fixed bias, voltage divider bias and Collector to base bias, bias stabilization for fixed bias and voltage divider bias, Stability factor S(β)		
	2.3	BJT configurations: CE, CB, CC		
	2.4	BJT as a switch		
3	**Field Effect Transistors (FET)**		09	CO2
	3.1	MOSFET: Structure, working, characteristics, two terminal basic MOS capacitor structure, nonideal current voltage characteristics: finite output resistance, Body effect		
	3.2	DC circuit analysis, load line and modes of operation and design of biasing circuits of MOSFET		
	3.3	MOSFET configurations: Common Source		
	3.4	MOSFET as a switch		
4	**BJT Amplifiers**		12	CO3, CO4
	4.1	Bipolar linear amplifier, small signal Hybrid π equivalent circuit of BJT		
	4.2	Small signal mid-frequency equivalent circuit - hybrid-pi model		
	4.3	Small signal mid-frequency analysis of CE, CB and CC amplifier using hybrid-pi model for voltage gain, input and output impedance		
5	**FET Amplifiers**		08	CO3, CO4
	5.1	Load lines and small signal parameters, Small signal equivalent circuit		
	5.2	Analysis of Enhancement MOSFET CS, CG, CD amplifier using hybrid-pi model for voltage gain, input and output impedance		
		Total	45	

Reference Books

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/ Year
1	D. A. Neamen	*Electronic Circuit Analysis and Design*	McGraw Hill Education, India	3rd Edition 2014
2	Boylestad and Nashelesky	*Electronic Devices and Circuits Theory*	Pearson Education, India	10th Edition 2009
3	Thomas L. Floyd	*Electronic Devices*	Prentice Hall (Pearson)	10th Edition 2016
4	Adel S. Sedra, Kenneth C. Smith, Arun N Chandorkar	*Microelectronic Circuits Theory and Applications*	OXFORD International Edition,India	6th Edition, 2013

Course Code	Name of the Course					
216U03C303	Digital Logic Design					
Teaching Scheme	TH	P		TUT		Total
	03	-		-		03
Credits Assigned	03	-		-		03
Evaluation Scheme	Marks					
	LAB/TUT CA	CA (TH)		ESE	Total	
		IA	ISE			
	-	20	30	50	100	

Course pre-requisites:

- **NIL**

Course Objectives:

Digital systems have a prominent role in everyday life. To understand the operation of digital systems, it is necessary to have a basic knowledge of digital circuits and their logical functions.The objective of this course is to familiarize the student with fundamental principles of digital design. It provides coverage of classical hardware design for both combinational and sequential logic circuits.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Understand fundamentals of number systems ,codes and arithmetic operations.
CO2. Minimize logic functions using different techniques and represent using logic gates.
CO3. Implement combinational logic circuits and their applications using various devices.
CO4. Design and implement sequential logic circuits and their applications.
CO5. Understand principle of data converters.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1		**Fundamentals of digital design**	05	CO1
	1.1	**Number systems and codes**: Introduction to digital systems, binary, octal, hexadecimal number systems, signed and unsigned numbers BCD code, Gray code, weighted and non-weighted codes, self-complementary codes and Error detecting code		
	1.2	**Binary arithmetic**-Addition, subtraction using 1's and 2's complement, multiplication, division and BCD Arithmetic		
2		**Combinational Logic Circuits**	18	CO2, CO3
	2.1	**Logic Gates**: Review of Basic logic Gates, universal gates, Minimization of logical expression using Boolean Functions. Combinational logic representation using truth table, and standard SOP, POS form		
	2.2	**Minimization techniques**: Use of Boolean theorem, K- map up to five variables, Quine-McClusky method, realization using standard and universal gates		
	2.3	**Arithmetic circuits**: Adder, subtractor, controlled adder/ subtractor, BCD adder		
	2.4	**Use Of MSI Devices**: Multiplexer, de-multiplexer, decoder, encoder, comparator, multiplexer tree and decoder tree. Boolean functions implementation using above devices.		
	2.5	**Logic Families:** Introduction to logic families, characteristics of digital ICs, transfer characteristics and comparison of TTLand CMOS families.		
	2.6	**Programmable logic devices**: Introduction, realization of logic expressions with PROM, PLA and PAL		
		# Self-learning: priority encoder, 7 segment decoders		
3		**Sequential Logic Circuits**	10	CO4
	3.1	**Flip flops (FF):** SR, JK, T, D and master slave flip flops, Truth table and excitation table characteristics equation, conversion of flip flops		
	3.2	**Counter:** Asynchronous and synchronous counter, Design of mod counters, Timing diagram		
	3.3	**Shift registers:** shift left and shift right Registers, bi-directional shift register, applications, ring counter, Johnson counter		
		# Self-learning: Designing with MSI devices (ICs 7490,7492, 74163,74194 etc.)		

4	**Finite state Machines**		**08**	**CO4**
	4.1	**Finite state machines**: Moore and Mealy machine, block diagram, state diagram, analysis of clocked synchronous state machine		
	4.2	**Synchronous state machine design**: P.S –N.S. table, reduction using implication chart method, state assignments, design using flip flops, up/down counter, skipping state counter, sequence detector and simple FSM design problems		
5	**Data Convertors**		**04**	**CO5**
	5.1	**Data converters:** A to D Conversion techniques, Dual slope ADC, Ramp ADC, Successive approximation ADC D to A Conversion techniques- R - 2R ladder network		
		Total	**45**	

Self Learning: Students should prepare all self-learning topics on their own. Self learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in IA and or laboratory experiments.

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Morris Mano andMichael D. Ciletti	*Digital Design: With an Introduction to Verilog HDL*	Pearson Education, India	5th Edition 2013
2	A.P. Malvino and D.P.Leach	*Digital Principles and Applications*	McGraw-Hill Education, India	10th Edition 2015
3	R.P.Jain	*Modern Digital Electronics*	Tata McGraw Hill Education, India	4th Edition 2015
4	John M Yarbrough	*Digital logic: Applications and design*	Thomson Brooks/ Cole, India	India Edition2006

<table>
<tr><th>Course Code</th><th colspan="6">Name of the Course</th></tr>
<tr><td>216U03C304</td><td colspan="6">Microprocessor and Microcontroller</td></tr>
<tr><td colspan="7"></td></tr>
<tr><td rowspan="2">Teaching Scheme</td><td>TH</td><td colspan="2">P</td><td colspan="2">TUT</td><td>Total</td></tr>
<tr><td>03</td><td colspan="2">0</td><td colspan="2">0</td><td>03</td></tr>
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<tr><td colspan="7"></td></tr>
<tr><td rowspan="4">Evaluation Scheme</td><td colspan="6">Marks</td></tr>
<tr><td rowspan="2">LAB/TUT CA</td><td colspan="2">CA (TH)</td><td rowspan="2">ESE</td><td rowspan="2" colspan="2">Total</td></tr>
<tr><td>IA</td><td>ISE</td></tr>
<tr><td></td><td>20</td><td>30</td><td>50</td><td colspan="2">100</td></tr>
</table>

Course pre-requisites:

Fundamentals of Digital Electronics, Binary and Hexadecimal number systems.

Course Objectives:

Microprocessors and microcontrollers are the integral part of any embedded systems. The course will enable students to learn basic programming of Microprocessor, Microcontroller and essential fundamentals for interfacing different peripheral devices to develop applications.

Course Outcomes (CO):

After successful completion of the course the student will be able to:

CO1. Understand architecture of 8086
CO2. Understand architecture and memory organization of 8051
CO3. Write assembly language programs for 8086 and 8051
CO4. Understand Timer operations ,serial communication and their programming
CO5. Interface peripherals like ADC, DAC, Sensors etc. with microcontroller and develop various applications

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	**Microprocessor 8086 architecture**		10	CO1
	1.1	Introduction to microprocessors and microcontrollers , evolution of microprocessors		
	1.2	Architecture, organization, Memory segmentation, pin configuration of 8086 , Memory banking		
	1.3	Minimum mode and Maximum mode of 8086		
	1.4	Read and Write bus cycle		
	1.5	Interrupt structure of 8086		
2	**Instruction set and programming of 8086**		10	CO3
	2.1	Opcode, operands, instruction word bytes, instruction cycle, fetch operation, execute operation and machine cycle		
	2.2	Addressing modes: Register, direct, register indirect, immediate, based, indexed, based indexed, based indexed and displacement		
	2.3	Instruction Set of 8086: Classification and description of instructions		
	2.4	Assembly Language programming for 8086		
3	**8051 Microcontroller**		10	CO2
	3.1	Overview, features and memory organization of 8051		
	3.2	Architecture, Pin diagram, addressing modes and assembler directives		
	3.3	Instruction set of 8051: Arithmetic, logical data transfer, branching and Boolean		
	3.4	Assembly language programming for 8051		
`		#Self-learning topic: Programs using subroutine		
4	**Timers, Serial Communication and Interrupts in 8051**		10	CO4
	4.1	Timers in 8051, timer modes and programming in assembly language		
	4.2	Basics of serial communication, 8051 connection to RS232, serial data transfer mode and serial port programming in assembly language		
	4.3	Interrupt structure in 8051, programming timer, external hardware and serial communication interrupts		
5	**8051 peripheral interfacing**		05	CO5
	5.1	Interfacing of ADC-0808 with 8051		
	5.2	Interfacing of DAC-0808 with 8051		
	5.3	Interfacing of 14-pin LCD with 8051		
	5.4	Interfacing of 4X4 matrix Keyboard with 8051		
		#Self-learning topic: Interfacing of Bluetooth module and different Sensors		
		Total	45	

 # Self Learning: Students should prepare all self-learning topics on their own. Self learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in IA and or laboratory experiments

Reference Books

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/ Year
1	Kenneth J. Ayala	*The 8086 Microprocessor: Programming & Interfacing The PC*	Cengage Learning, India	3 rd Edition, 2008
2	Douglas V Hall	*Microprocessors and Interfacing*	Tata McGraw Hill Education, India	2 nd Edition, 2007
3	Muhammad Ali Mazidi Janice Gillispie Mazidi Rolin D. McKinlay	*The 8051 Microcontroller and Embedded Systems*	Pearson, India	2 nd Edition, 2008
4	Kenneth J. Ayala	*The 8051 Microcontroller*	Cengage Learning,India	3 rd Edition, 2004

Course Code	Name of the Course			
216U03C305	Electrical Networks			
Teaching Scheme	TH	P	TUT	Total
	03	-	01	04
Credits Assigned	03	-	01	04

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	25	20	30	50	125

Course pre-requisites:

- Elements of Electrical and Electronics Engineering
- Partial fraction expansion, matrices, Laplace transforms and differential equations

Course Objectives:

Electrical Networks find its applications in various allied disciplines of Electronics Engineering and Telecommunication Engineering. This course will help the students to learn the fundamental concepts of mesh and nodal analysis, network theorems and functions, transient analysis, two port analysis, synthesis of realizable circuits from transfer functions and passive filters.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Analyse DC and AC circuits using mesh analysis, nodal analysis and network theorems.
CO2. Analyse transient and steady state response in time and frequency domain.
CO3. Determine two port network parameters.
CO4. Synthesize circuits from transfer functions.
CO5. Learn and design different types of filters and their parameters

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1		**Network Analysis**	14	CO1
	1.1	DC with Dependant Sources: Mesh and nodal analysis, Superposition theorem, Thevenin's, Norton's Theorem and Maximum Power Transfer Theorem		
	1.2	AC Circuits: Mesh and nodal analysis, Superposition theorem , Thevenin's theorem, Norton's Theorem and Maximum Power Transfer Theorem		
	1.3	Coupled Circuits: Self and mutual inductance, Coefficient of coupling, equivalent inductances and mesh analysis using dot convention.		
2		**Transient response of R-L, R-C, R-L-C circuits (Series and Parallel combination)**	10	CO2
	2.1	Test inputs for transient response: step, impulse and ramp input.		
	2.2	Initial conditions		
	2.3	Solution based on time domain analysis		
	2.4	Solution based on Laplace transform		
3		**Two port networks (Z,Y, ABCD and Hybrid Parameters)**	08	CO3
	3.1	Driving point and transfer function		
	3.2	Concept of reciprocity and symmetry		
	3.3	Two port network parameters and their interrelation		
	3.4	Interconnection of two port networks: Cascade connection in terms of ABCD parameters.		
4		**Network functions and Synthesis**	08	CO4
	4.1	Poles and Zeros of network functions and their significance		
	4.2	Time domain response using pole zero plot		
	4.3	Hurwitz polynomial		
	4.4	Positive real functions		
	4.5	Properties and Synthesis of LC circuit using Foster I and II, Cauer I and II		
5		**Passive Filters**	05	CO5
	5.1	Classification of Filters: Low pass, high pass, band pass and band stop filters, transfer functions, frequency response and filter parameters.		
	5.2	T and pi network		
	5.3	Design of constant k filters: Low pass, high pass, band pass and band stop filters		
	5.4	Design of constant m-derived filters: Low pass, high pass, band pass and band stop filters		
		Total	**45**	

Reference Books

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Ravish Singh	*Network Analysis and Synthesis*	McGraw Hill Education (India) Private Limited.	Seventh Reprint 2018
2	D. Roy Choudhury	*Networks and Systems*	New Age International Publishers.	2nd Edition 2010
3	A. Sudhakar and Shyammohan. S. Palli	*Circuits and Networks*	McGraw Hill Education (India) Private Limited.	4th Edition 2010
4	C. K. Alexander and M. N. O. Sadiku	*Electric Circuits*	McGraw Hill Education (India) Private Limited.	5th Edition 2013
5	Nilsson Riedel	*Electric Circuits*	Pearson Education India.	11th Edition 2018

Instructions:	• TUT includes journal / written record of tutorial .Tutorials will be evaluated out of **25 marks each** A minimum of 8-10 tutorials will be performed in a semester. Average of all tutorials will be considered as final TUT CA marks.

<table>
<tr><th>Course Code</th><th colspan="5">Name of the Course</th></tr>
<tr><td>216U06I306</td><td colspan="5">Foundational Course in Indian Knowledge Systems</td></tr>
<tr><td colspan="6"></td></tr>
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<tr><td rowspan="4">Evaluation Scheme</td><td colspan="5">Marks</td></tr>
<tr><td rowspan="2">LAB/TUT CA</td><td colspan="2">CA (TH)</td><td rowspan="2">ESE</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>ISE</td></tr>
<tr><td>--</td><td>50</td><td>--</td><td>50</td><td>100</td></tr>
</table>

Course Objectives:

1. to introduce students to the rich diversity of Indian knowledge systems
2. to introduce the life and works of important figures in the respective domains
3. to explore the underlying philosophical and cultural ethos that distinguishes Indian Knowledge Systems
4. to emphasise continuity of the tradition into modern times, wherever applicable.

Course Outcomes

At the end of successful completion of the course the student will

CO1. have a clear understanding of the different domains of Indian Knowledge Systems
CO2. have become aware of the contribution of great figures in the respective fields
CO3. have an understanding of how culture impacts creation of knowledge
CO4. learn to investigate correlations and synthesis leading to development of any knowledge system

Detailed Syllabus

Module No.	Unit No.	Topics	Hours per topic	Hours per Unit
1		**Sources of Indian Knowledge Systems**		**4**
	1.1	IKS - Concept, scope, relevance to our world today.	1	
	1.2	Textual sources, historical accounts, archaeological evidence, inscriptions, coins etc	3	
2		Why study IKS?		**2**
	2.1	Importance of the IKS, its interconnections and relevance to the modern fields of science.	2	
3		**Yoga: Basic Practices and Philosophy**		**6**
	2.1	Maharshi Patanjali, Swami Satyananda Saraswati, B K S Iyengar, Swami Kuvalayananda, Sri Yogendra	2	
	2.2	Body loosening exercises Importance of breath, developing concentration Yoga for mind-body wellness	4	
4		**Genres of Ancient Literature**		**6**
	4.1	Religious: Vedic texts, Buddhist and Jain texts;	2	
	4.2	Epics, Puranas, Sangam literature	2	
	4.3	Poetry, Mathematics, and Scientific Literature	2	
5		**Leadership and Ethical Values**		**6**
	5.1	Selections from Shantiparva of Mahabharata, Arthasastra, Panchatantra, Hitopadesha, Jataka tales, Bhagavadgita, Dhammapada and Thirukkural: Discussions on ethical values	3	
	5.2	Leadership qualities as reflected through ancient Indian literature: Lessons for modern leadership challenges	3	
6	6.1	**Art:** **Sculpture(iconography) and Paintings**	3	
		Iconography: Ellora (Buddhist and Jain) and Hampi (Hindu)		
		Paintings: Ajanta(Buddhist), Ellora(Jain), Brihadeshvar Temple-Thanjavur.(Hindu)		
	6.2	**Architecture:** **Rock-cut caves and Temple Architecture**	3	
		Rock-cut caves: Kanheri, Elephanta, Ellora (any two sites can be used for detailed discussion)		

		Temple architecture: Pattadakal, Konark Temple, Jagannatha Temple-Puri, Bodh Gaya, Dilwara Temple-Mount Abu (any two sites can be used for detailed discussion)		
7		**Ancient Indian Mathematics**		**6**
	7.1	Shulba Sutras, Bakshali Manuscript	2	
	7.2	Aryabhatiya: place value system, approximation of the value of π, geometry	2	
	7.3	Bhaskaracharya: different approach to teaching mathematics	2	
8		**Ancient Indian Astronomy**		**6**
	8.1	Indian calendar system: Sayana-nirayana calendar, Panchanga	3	
	8.2	Spherical trigonometry, Eclipse computation	3	
9		**Ancient Indian Agriculture**		**6**
	9.1	General management of Agriculture and Farming Operations	3	
	9.2	Cattle Management, Weather predictions	3	
10		**Trade and Commerce**		**6**
	10.1	Silk route, Uttarapatha and Dakshinapatha, Maritime route	3	
	10.2	Barter system, Numismatics	3	
11		**Ancient Indian Society**		**6**
	11.1	Law and Justice	3	
	11.2	Marriage Laws, Inheritance	3	
12		**Chemistry and Metallurgy**		**6**
	12.1	Multiple sources such as archaeological artifacts, temple icons,	1	
	12.2	Metals and beads	2	
	12.3	Chemistry of dyes, Colouring materials	2	
	12.4	Paintings and Painting materials	1	
		Total Hours		**30***

* The first two modules remain the core and other modules can be selected (any 4 modules from module 3 to module 12) by the college depending upon the availability of the teachers making it to a 30hrs course.

Recommended Books:

Text Book on IKS:

1. Mahadevan B., Bhat Vinayak Rajat, Nagendra Pavana R. N. Introduction to Indian Knowledge System: concepts and Applications, PHI Learning Pvt. Ltd. 2022

Reference Books:

1. Amma Sarasvati T. A., Geometry in Ancient and Medieval India, MLBD, Delhi, 1st ed. 1999, reprint 2007
2. Acharya, P. K., Indian Architecture According to ManasaraShilapshastra, Oxford University Press 1927
3. Altekar, A.S., Education in Ancient India, Gyan Books, 2010
4. Appleton Naomi, Jataka Stories in Theravada Buddhism: Narrating the Bodhisatta Path, Routledge Publication, New York 2016
5. Bhatacharyya, T. , Study of Vastuvidya or Canon of Indian Architecture, Patna 1976
6. Bose, N. K., Orissan temple Temple Architecture (Vastushastra) [With Sanskrit text and English translation), Bharatiya Kala Prakashana, Delhi 20017
7. Chatterjee, Satischandra & Datta, Dhirendra Mohan. An introduction to Indian Philosophy, Rupa Publications India Pvt. Ltd., New Delhi, 7th edition, 1968
8. Clark Walter Eugene, The Aryabhatiya of Aryabhata- An Ancient Indian Work On Mathematics and Astronomy, Delta Book World, India, 2021
9. Coomaraswamy, Ananda K. Early Indian Architecture: Cities and City-Gates, Munshiram Manoharlal Publishers, 2002
10. D M Bose, S N Sen and B V Subbarayappa, eds; A Concise History of Science in India, INSA; 2009
11. Datta Bibhutibhushan & Singh Avadhesh Narayan, History of Hindu Mathematics, 1935, repr. Bharatiya Kala Prakashan, Delhi, 2004
12. Datta Bibhutibhushan, Ancient Hindu Geometry: The Science of the Śulba, 1932, reprint. Cosmo Publications, New Delhi, 1993
13. Deglurkar, G. B, Temple Architecture and Sculpture of Maharashtra, Nagpur University, Nagpur 1974
14. Dehejia, Vidya, Early Buddhist Rock Temples A Chronological Study, London, 1972
15. Dehejia, Vidya, Early Stone Temples of Orissa, Vikas Publishing House, Delhi 1979
16. Divakaran P. P., The Mathematics of India: Concepts, Methods, Connections, Hindustan Book Agency, 2018
17. Dr. Mishra Shiv Shekhar, Fine Arts & Technical Sciences in Ancient India with special reference to Someśvara's Mānasollāsa; Krishnadas Academy, Varanasi 1982
18. Ed. and Trs. Majumdar Girija Prasanna, Banerji Sures Chandra, Krisi-Parasara, Asiatic Society, Kolkata, 1960
19. Ed. Tr. Kangale, R. P, Kautiliya Arthashastra, University of Bombay, Bombay, 1960
20. Gupta, Swarajya Prakash, Asthana Shashi, Elements of Indian Art: Including Temple Architecture, Iconography & Iconometry, Indraprastha Museum of Art and Archeology, 2007
21. Kane P.V., History of Sanskrit Poetics, Motilal Banarasidass, New Delhi, 4th edition, 1971

22. Larson, G. J. (Ed.) and Bhattacharya, R. (Ed.) , Encyclopaedia of Indian Philosophies: Yoga: India's Philosophy of Meditation, Vol. XII, Motilal Banarasidas Publishers Pvt. Ltd., Delhi, 1st edi., 2008
23. Paranjpe Kalpana, Ancient Indian insights and Modern Science: A Rare Book, Bhandarkar Oriental Research Institute, Pune, 2022
24. Radhakrishnan, S., The Principal Upanisads, Oxford University Press, Delhi, 1992
25. Rahman A., Alvi M. AKhan .S A., Ghori, Murthy Samba K. V., Science and Technology in Medieval India - A Bibliography of Source Materials in Sanskrit, Arabic and Persian, 1982
26. Rao Balachandra S., Indian Astronomy – An Introduction, Universities Press (India) Limited, Hyderabad, 2000

<table>
<tr><th>Course Code</th><th colspan="4">Name of the Course</th></tr>
<tr><td>216U03L301</td><td colspan="4">Data Structures and Analysis of Algorithms Laboratory</td></tr>
<tr><td colspan="5"></td></tr>
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<tr><td>01</td><td>02</td><td>-</td><td>03</td></tr>
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<table>
<tr><td rowspan="4">Evaluation Scheme</td><td colspan="5">Marks</td></tr>
<tr><td rowspan="2">LAB/TUT CA</td><td colspan="2">CA (TH)</td><td rowspan="2">ESE</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>ISE</td></tr>
<tr><td>75*</td><td>-</td><td>-</td><td>-</td><td>75</td></tr>
</table>

Course pre-requisites:

- Basic concepts of C programming

Course Objectives:

The data structure and analysis of algorithm provide a set of techniques to the programmer for handling the data efficiently. The objective of the course is to design, write, and analyse the performance of C programs that handle structured data and perform more complex tasks. Course also introduces the concept of data representation & manipulation with a view for efficiency, maintainability, and code-reuse.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Explain the different data structures used in problem solving
CO2. Understand and analyse the time and space complexity of an algorithm
CO3. Demonstrate sorting and searching methods
CO4. Understand data structure operations and fundamental algorithms
CO5. Write C and PYTHON programs to implement data structure algorithms

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	Introduction		2(T), 4(P)	CO1, CO2
	1.1	Basic Data Structures: Data Structures, Types of Data Structures, Abstract data type (ADT), Data Structures in C		
	1.2	Analysis of Algorithms: Rates of Growth: O(n), Ω(n), Θ(n), o(n), ω(n), Time Complexity, Space Complexity		
2	**Linear Data Structures : Linked Lists**		4(T), 8(P)	CO4, CO5
	2.1	Linked list as ADT, Memory allocation & De-allocation for a Linked List, Linked List operations, Types of Linked List		
	2.2	Implementation of Singly Linked List, Circular linked list, doubly linked list		
3	**Linear Data Structures : Stack and Queue**		3(T), 6(P)	CO4, CO5
	3.1	**Stack:** The Stack as an ADT, Stack operations, Array Representation of Stack, Linked Representation of Stack.		
	3.2	**Queue**: The Queue as an ADT, Queue operation, Array Representation of Queue, Linked Representation of Queue, and Circular Queue.		
4	**Non-linear Data Structures**		3(T), 6(P)	CO4, CO5
	4.1	**Tree** :Basic trees concept, Binary tree representation, Binary tree operations, Binary tree traversal, Binary search tree implementation.		
	4.2	**Graph**: Basic concepts, Graph Representation, Graph traversal-Depth First Search & Breadth First Search.		
5	**Search and sort algorithms**		3(T), 6(P)	CO2, CO3
	5.1	**Sorting**: Sort Concept, Bubble Sort, Insertion Sort, Merge Sort		
	5.2	**Searching**: Linear Index Search, Binary Search		
		Total	**45**	

Reference Books

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Langsam Yedidyah, Augenstein J Moshe, Tenenbaum M Aaron	*DATA STRUCTURES USING C AND C++*	PHI Learning	2nd Edition, 2009
2	Richard Gilberg, Behrouz A. Forouzan	*Data Structures: A Pseudocode Approach with C*	Cengage Learning	2^{nd} Edition 2004
3	Ellis Horowits, Sartaj Sahni, and Susan Anderson-Freed	*Fundamentals of data structures in C*	University Press	2^{nd} Edition 2008
4	ReemaThareja	*Data Structure and Algorithm*	OXFORD	2^{nd} Edition , 2014

*Distribution of marks: 25 marks term work, 50 marks lab IA work.

Instructions:	● LAB CA includes journal / written record of experiments (25 marks), Internal assessment (50 marks) ● The course faculty will determine the Internal Assessment (IA) tasks assigned to students.

<table>
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<tr><td>216U03L302</td><td colspan="5">Basic Electronics Circuits Laboratory</td></tr>
<tr><td colspan="6"></td></tr>
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<tr><td rowspan="4">Evaluation Scheme</td><td colspan="5">Marks</td></tr>
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<tr><td>IA</td><td>ISE</td></tr>
<tr><td>50</td><td>--</td><td>--</td><td>--</td><td>50</td></tr>
</table>

Instructions:	- LAB CA includes journal / written record of experiments (25 marks) and oral based on experiments performed (25 marks). - Experiments will be evaluated out of **25 marks each**. A minimum of 8-10 will be performed in a semester. - Oral will be taken at the end of the semester related to all experiments for 25 marks. **The total marks for LAB CA will be 50**.

<table>
<tr><th>Course Code</th><th colspan="5">Name of the Course</th></tr>
<tr><td>216U03L303</td><td colspan="5">Digital Logic Design Laboratory</td></tr>
<tr><td colspan="6"></td></tr>
<tr><td rowspan="2">Teaching Scheme (Hrs./Week)</td><td>TH</td><td colspan="2">P</td><td>TUT</td><td>Total</td></tr>
<tr><td>--</td><td colspan="2">02</td><td>--</td><td>02</td></tr>
<tr><td>Credits Assigned</td><td>--</td><td colspan="2">01</td><td>--</td><td>01</td></tr>
<tr><td colspan="6"></td></tr>
<tr><td rowspan="4">Evaluation Scheme</td><td colspan="5">Marks</td></tr>
<tr><td rowspan="2">LAB/TUT CA</td><td colspan="2">CA (TH)</td><td rowspan="2">ESE</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>ISE</td></tr>
<tr><td>50</td><td>--</td><td>--</td><td>--</td><td>50</td></tr>
</table>

Instructions:	• LAB CA includes journal / written record of experiments (25 marks) and Evaluations based on experiments performed (25 marks).The nature and schedule of the evaluations will be informed by concerned faculty member. • Experiments will be evaluated out of **25 marks each**. A minimum of 8-10 experiments will be performed in a semester based on syllabus of course Digital Logic Design • **The total marks for LAB CA will be 50**.

Course Code	Name of the Course				
216U03L304	Microprocessor and Microcontroller laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	--	02	--	02	
Credits Assigned	--	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

*Distribution of marks:

Instructions:	• LAB CA includes journal / written record of experiments (25 marks) and oral based on experiments performed (25 marks). • Experiments will be evaluated out of **25 marks each**. A minimum of 8-10 will be performed in a semester. • Practical and oral will be taken at the end of the semester related to all experiments for 25 marks. **The total marks for LAB CA will be 50**.

SEMESTER IV

<table>
<tr><th>Course Code</th><th colspan="5">Name of the Course</th></tr>
<tr><td>216U03C401</td><td colspan="5">Probability, Statistics and Optimization Techniques</td></tr>
<tr><td colspan="6"></td></tr>
<tr><td rowspan="2">Teaching Scheme</td><td>TH</td><td>P</td><td colspan="2">TUT</td><td>Total</td></tr>
<tr><td>03</td><td>-</td><td colspan="2">-</td><td>03</td></tr>
<tr><td>Credits Assigned</td><td>03</td><td>-</td><td colspan="2">-</td><td>03</td></tr>
<tr><td colspan="6"></td></tr>
<tr><td rowspan="4">Evaluation Scheme</td><td colspan="5">Marks</td></tr>
<tr><td rowspan="2">LAB/TUT CA</td><td colspan="2">CA (TH)</td><td rowspan="2">ESE</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>ISE</td></tr>
<tr><td>-</td><td>20</td><td>30</td><td>50</td><td>100</td></tr>
</table>

Course pre-requisites:

- Basics of Linear Algebra and Calculus

Course Objectives:

The main objective of the course is to provide education to students so that they can meet the growing need for qualified personnel in the fields of data science. The course content is designed to develop theoretical and practical literacy in computer science and statistics, with relevance in data science, so that students are able to critically select and apply appropriate methods and techniques to extract relevant and important information from data.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Apply concepts of continuous and discrete distributions to solve Engineering problems.
CO2. Learn statistical concepts for data summarization and data visualization.
CO3. Understand sampling and parameter estimation for statistical analysis.
CO4. Conduct hypothesis tests including Z-test, t-test, and F-test and use them in Statistical Model development.
CO5. Learn and apply Regression and Optimization approaches for problem solving.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1		**Probability**		
	1.1	Conditional and Joint Probability, Bayes' Theorem, PDF & CDF of Continuous and Discrete random variable, Binomial Distribution, Poisson Distribution, Geometric Distribution, Uniform Distribution, Exponential Distribution, Normal Distribution, Chi-Square Distribution, Student's t-Distribution, F-Distribution	04	CO1
2		**Statistical Data Analysis**	08	CO2
	2.1	Data types and Data pre-processing techniques		
	2.2	Data summarization using Descriptive Statistics, Mean, Arithmetic Mean, Geometric Mean, Harmonic Mean, Median, Skewness, Kurtosis, Mode, Range, Variance, Standard Deviation, Percentile, IQR, MAD, Correlation, Covariance		
	2.3	Data visualization as a analytical tool using basic graphs, Histogram, Bar plot, Line plot, Scatter plot, Box plot, Stem and Leaf plot, Pie chart, Heat map, Density and Contour plots, Data visualization as a medium of communication		
3		**Sampling and Estimation**	12	CO3
	3.2	Random variable, Statistical Inference, Population Parameter and Sample Statistic. Sampling Distribution, Sampling distribution of mean and variance, Central Limit Theorem		
	3.3	Population Parameter Estimation: Introduction, Maximum Likelihood Estimation(MLE) method for point estimation of population parameter, MLE to estimate parameter of Binomial, Poisson, Gaussian and Uniform distribution		
	3.4	Confidence Interval method for population parameter estimation: Two sided and One sided confidence Interval estimates, For a Normal mean when variance of population is known, For a Normal mean when variance of population is not known , For variance of Normal distribution		
4		**Hypothesis testing**	06	CO4
	4.1	Introduction, Setting up a hypothesis test, Null Hypothesis and Alternative Hypothesis, One tailed and two tailed test, Type I and type II errors		
	4.2	Hypothesis testing for Population Mean with known variance: z-test, Hypothesis testing for Population Mean under unknown variance: t-test		
	4.3	ANOVA: Introduction, One-way ANOVA, Two-way ANOVA		

5	**Regression and Optimization**			
	5.1	Regression: Supervised learning, Linear Regression, Simple Linear Regression (SLR), Estimation of SLR model parameters using Ordinary Least Squares (OLS) method; Multiple Linear Regression (MLR) model, Estimation of model parameters using OLS Method, Multi-collinearity; Interpretation of SLR and MLR model parameters, model assessment using model parameters, Validation of assumptions in model using diagnostic plots; Regression Model building for data consisting of quantitative and qualitative variables	**15**	**CO5**
	5.2	Optimization basics; Optimization for convex functions, Gradient Descent method; Optimization for non-differentiable functions, subgradient descent method; Newton's method for second order differentiable functions; Constrained optimization using Lagrangian method		
		Total	**45**	

Reference Books*

Sr. No.	**Name/s of Author/s**	**Title of Book**	**Publisher**	**Edition/Year**
1	Sheldon M. Ross	*Probability and Statistics for Engineers and Scientists*	Academic Press	4th edition, 2014
2	Jiawei Han, Micheline Kamber, Jian Pei	*Data Mining: Concepts and Techniques*	Morgan Kaufmann	3rd edition 25 July 2011
3	Jake VanderPlas	*Python Data Science Handbook: Essential Tools for Working with Data*	O'Reilly	1st edition, January 2016
4	Howard J. Seltman	*Experimental Design and Analysis*		July 2018
5	Douglas C. Montgomery, Elizabeth A. Peck, G. Geoffrey Vining	*Introduction to Linear Regression Analysis*	WILEY	6th edition, February 2021
6	J. K. Sharma	*Operation research: Theory and Applications*	Laxmi Publications, India	6th Edition 2017

Course Code	Name of the Course			
216U03C402	**Analog Electronics**			
Teaching Scheme	**TH**	**P**	**TUT**	**Total**
	03	**-**	**-**	**03**
Credits Assigned	**03**	**-**	**-**	**03**

Evaluation Scheme	**Marks**				
	LAB/TUT CA	**CA (TH)**		**ESE**	**Total**
		IA	**ISE**		
		20	**30**	**50**	**100**

Course pre-requisites:

- Elements of Electrical and Electronics Engineering
- Basic Electronic Circuits

Course Objectives:

The objective of this course is to explore analysis and design of Analog Electronic Circuits- multistage amplifier, differential amplifier, operational amplifiers, oscillators and power amplifiers.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to
CO1. Analyse the frequency response of BJT and FET amplifier.
CO2. Understand the basics of Analog Integrated Circuits
CO3. Understand working and design of BJT power amplifiers
CO4. Explore oscillator circuits using operational amplifiers

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	**Frequency Response of BJT and MOSFET Amplifiers**		16	CO1
	1.1	High frequency hybrid-pi model of BJT and MOSFET, Miller effect and capacitance and gain bandwidth product		
	1.2	Effect of coupling, bypass and junction capacitors on frequency response, low and high frequency response of single stage BJT - CE, CB, CC and MOSFET - CS, CG, CD amplifiers		
	1.3	Mid-frequency analysis of multistage - CS-CS and Cascode CS- CG amplifiers and Darlington pair, design of two stage amplifiers		
	1.4	Low and high frequency response of multistage - CS-CS and Cascode CS-CG amplifiers		
2	**Biasing Techniques for Integrated Circuits**		06	CO2
	2.1	BJT-MOSFET two and three transistor current sources		
	2.2	Wilson and Widlar current sources		
3	**Fundamentals of Operational Amplifier**		09	CO2
	3.1	Practical Op-Amp parameters, Finite open loop gain, frequency response, offset voltage, input bias current, CMRR		
	3.2	Op-amp with negative feedback, block diagram representation of feedback Configurations: Voltage-Series, Voltage-shunt, Current-Series, Current-shunt. Analysis of Voltage-Series Feedback Amplifier		
	3.3	Analysis and design of Adder, Subtractor, Active filters, V to I converters, Schmitt Trigger, Voltage Comparators, Logarithmic amplifiers, Sample and Hold circuit, Peak detector		
4	**BJT Power amplifier**		09	CO3
	4.1	Amplifier classes, Series-Fed Class A, Transformer coupled Class A amplifier, Class B, Class C, Class D, Class AB amplifiers		
	4.2	DC analysis, Power efficiency, Distortion and Heat Sink		
	4.3	Design of class A power amplifier		
5	**Oscillator Circuits**		05	CO4
	5.1	Concept of positive feedback, conditions for oscillations and Barkhausen Criteria for sustained oscillations		
	5.2	Hartley oscillator, Crystal oscillator, Phase shift oscillator, Wien Bridge oscillator		

Total	**45**	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/ Year
1.	Donald A. Neamen	*Electronic Circuit Analysis and Design*	Tata McGraw Hill, India	3rd Edition 2014
2.	Adel S. Sedra, Kenneth C. Smith, Arun N Chandorkar	*Microelectronic Circuits Theory and Applications*	OXFORD International Edition,India	6th Edition, 2013
3.	Robert Boylestad , Louis Nashelsky	*Electronic Devices and Circuit Theory*	Pearson Education,India	10th Edition 2015
4.	Sergio Franco	*Design with Operational Amplifiers and analog integrated circuits*	McGraw Hill International, New Delhi, India	3rd Edition 2017
5.	Ramakant A. Gayakwad	*Op-Amps and Linear Integrated Circuit*	Pearson Prentice Hall, India	4th Edition,2012

<table>
<tr><th>Course Code</th><th colspan="6">Name of the Course</th></tr>
<tr><td>216U03C403</td><td colspan="6">Communication Systems</td></tr>
<tr><td colspan="7"></td></tr>
<tr><td rowspan="2">Teaching Scheme</td><td>TH</td><td colspan="2">P</td><td colspan="2">TUT</td><td>Total</td></tr>
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<tr><td colspan="7"></td></tr>
<tr><td rowspan="4">Evaluation Scheme</td><td colspan="6">Marks</td></tr>
<tr><td rowspan="2">LAB/ TUT CA</td><td colspan="2">CA (TH)</td><td rowspan="2">ESE</td><td colspan="2" rowspan="2">Total</td></tr>
<tr><td>IA</td><td>ISE</td></tr>
<tr><td>50</td><td>20</td><td>30</td><td>50</td><td colspan="2">150</td></tr>
</table>

Course pre-requisites:

Basic Electronics Circuits

Course Objectives:

The objective of the course is to introduce basic principles and techniques used in analog and pulse communication systems. The course addresses various concepts related to analog and pulse communication systems such as modulation, demodulation, transmitter, receiver and noise effect. The course also introduces analytical techniques to evaluate the performance of communication systems in time and frequency domains.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Analyze and compare analog modulation schemes
CO2. Understand working of analog communication transmitter and receiver systems
CO3. Compare Pulse modulation systems and multiplexing techniques.
CO4. Understand noise effect on the performance of Communication Systems

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1		**Fundamentals of Electronic Communication Systems**	05	CO2 CO4
	1.1	Electromagnetic spectrum, Basic block diagram of communication system, Concept of modulation and demodulation		
	1.2	Sources of noise, signal to noise ratio, noise factor, noise figure, noise temperature and related numerical.		
2		**Amplitude Modulation and Demodulation**	11	CO1 CO2 CO4
	2.1	AM Signal: Waveform, mathematical representation, spectrum, modulation index, bandwidth, voltage distribution, power calculations and related numerical.		
	2.2	DSB-FC system: Modulation, low level and high level modulation; Demodulation: envelope detector, practical diode detector		
	2.3	DSB-SC system: Modulation, Balanced modulator, Ring modulator; Demodulation: Coherent Detector, Product detector		
	2.4	SSB system: Modulation: Filter method, Phase shift method, Third method; Pilot carrier SSB, Concept of VSB and ISB		
3		**Basic Features of Angle Modulation and Demodulation**	13	CO1 CO2 CO4
	3.1	Angle Modulation: Waveform, mathematical representation, modulation index, frequency and phase deviation, spectrum , bandwidth ; Narrow Band FM, Wide Band FM		
	3.2	FM system: Modulation, Direct methods –Varactor diode modulator, FET reactance modulator; Indirect method – Armstrong’s method; Noise triangle in FM; Pre-emphasis and de-emphasis; FM transmitter		
	3.3	PM system: Modulation, Direct and Indirect PM modulator; Relationship and comparison between FM and PM		
	3.4	FM Demodulation: Foster-Seeley discriminator, Ratio detector, FM demodulator using PLL; Comparison between AM, FM and PM		
4		**Radio Receivers**	06	CO2 CO4
	4.1	Characteristics of radio receiver, Choice of Intermediate Frequency , Image frequency, Concept of Automatic Gain Control and its type		
	4.2	Receivers: Super-heterodyne receiver DSB-FC, FM , Automatic Frequency Control		
5		**Pulse Modulation and Demodulation**	10	CO2 CO3 CO4
	5.1	Sampling theorem for low pass signals		
	5.2	PAM, PWM and PPM systems: Modulation and		

		demodulation, Applications		
	5.3	PCM: Transmitter and receiver; noise considerations in PCM		
	5.4	DM: Transmitter and receiver; noise considerations in DM; Adaptive delta modulation		
	5.5	Multiplexing: Time Division Multiplexing, Frequency Division Multiplexing		
		Total	**45**	

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	B.P. Lathi, Zhi Ding	*Modern Digital and Analog Communication system*	Oxford University Press	Fourth Edition,2017
2	Herbert Taub, Donald Schilling , Goutam Saha	*Principles of Communication Systems*	McGraw-Hill Electrical and Electronic Engineering Series	Fourth Edition,2013
3	Wayne Tomasi	*Electronics Communication Systems*	Pearson education	Fifth Edition, 2008
4	R.P. Singh and S.D. Sapre	*Communication systems Analog and Digital*	McGraw Hill, India	Second Edition, 2017
5	Gorge Kennedy Bernard Davis SRM Prasanna	*Electronic communication system*	McGraw-Hill	Fifth Edition, 2012

*In addition to printed books, faculty can suggest (authentic) urls or e-books, e-contents etc.

<table>
<tr><th>Course Code</th><th colspan="6">Name of the Course</th></tr>
<tr><td>216U03C404</td><td colspan="6">Signals and Systems</td></tr>
<tr><td colspan="7"></td></tr>
<tr><td rowspan="2">Teaching Scheme</td><td>TH</td><td colspan="2">P</td><td colspan="2">TUT</td><td>Total</td></tr>
<tr><td>03</td><td colspan="2">-</td><td colspan="2">01</td><td>04</td></tr>
<tr><td>Credits Assigned</td><td>03</td><td colspan="2">-</td><td colspan="2">01</td><td>04</td></tr>
<tr><td colspan="7"></td></tr>
<tr><td rowspan="4">Evaluation Scheme</td><td colspan="6">Marks</td></tr>
<tr><td rowspan="2">LAB/TUT CA</td><td colspan="2">CA (TH)</td><td rowspan="2">ESE</td><td rowspan="2" colspan="2">Total</td></tr>
<tr><td>IA</td><td>ISE</td></tr>
<tr><td>25</td><td>20</td><td>30</td><td>50</td><td colspan="2">125</td></tr>
</table>

Course pre-requisites:

- Basic calculus, Applied Mathematics-I, Applied Mathematics-II

Course Objective:

This course aims to introduce concept of signals and systems and their analysis both in continuous time and discrete time, in preparation for more advanced subjects in digital signal processing including audio, image and video processing, communication theory.

Course Outcomes (CO):

At the end of successful completion of the course students will be able to

CO1. Classify signals and perform basic operations on continuous and discrete time signals.
CO2. Classify systems and analyse continuous and discrete time Linear Time Invariant system in time domain.
CO3. Analyse discrete time Linear Time Invariant system using Z-Transform.
CO4. Represent signals and understand properties of CT systems in frequency domain
CO5. Represent signals and understand properties of DT systems in frequency domain.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1		**Basics of Signals and Systems**	10	CO1
	1.1	**Introduction to Signals and system** Explanation of a broad applicability of signals & system applied to mechanical systems, electrical systems, financial systems and biological systems.		
	1.2	**Classification of signals**: Continuous (CT) and Discrete time (DT) signals, Deterministic and Random Signals, Periodic and non-periodic, even and odd, energy and power signals, Causal and Non-casual signals.		
	1.3	**Elementary DT and CT signals** Unit Step signal, Unit Ramp signal, Unit impulse, exponential and sinusoidal signals. Operations on signals		
2		**Continuous Time and Discrete Time Systems**	10	CO2
	2.1	**Properties of DT and CT systems:** Linearity, causality, time variance, stability and static/dynamic		
	2.2	**Discrete Time LTI system:** Representation of discrete time signals in terms of impulses, Unit impulse response and convolution sum. Block diagram and difference equation representation of DT-systems (Solution of difference equations using classical method is not expected)		
	2.3	**Continuous Time LTI system:** Representation of continuous time signals in terms of impulses, Unit impulse response and convolution integral, Block diagram and differential equation representation of CT-systems (Solution of differential equations using classical method is not expected)		
3		**Z-Transform**	8	CO3
	3.1	Definition, Properties of Z-Transform (No derivations), Z-Transforms of standard signals, unilateral Z-Transform, Inverse Z-Transform.		
	3.2	**Analysis and Characterization of DT-LTI Systems using Z-Transform**: Relation of Unit Impulse Response and System Function, Response of DT system using Z-transform, properties of DT-LTI system in Z-domain.		
Self-study: derivations of important properties of Z-Transform, Concept map for DT-LTI system; Relation: DT system-Block diagram, difference equation, unit impulse response, system function				

4	**Frequency characterization of CT signals and LTI systems**		**9**	**CO4**
	4.1	**CT Fourier Series (CTFS):** Response of LTI systems to complex exponentials, Fourier series: Representing periodic signals in terms of Sinusoids and harmonics, Orthogonality, Trigonometric and Exponential of Fourier series, properties of CTFS (No derivations),filters described by differential equation.		
	4.2	**CT Fourier Transform (CTFT)**: Representing CT aperiodic signals by Fourier Transform – Definition of CTFT, CTFT of CT aperiodic signals, Properties of CTFT (No derivations). Inverse Fourier Transform, Magnitude and Phase response of CT-LTI systems.		
5	**Frequency characterization of DT Signals and LTI systems**		**8**	**CO5**
	5.1	**DT Fourier Series (DTFS):** Fourier Series representation of DT periodic Signals, properties of DTFS (No derivations), filters described by difference equations		
	5.2	**DT Fourier Transform (DTFT)**: Representing DT aperiodic signals by DT Fourier Transform – Definition of DTFT, Properties of DTFT (No derivations), Inverse DTFT, Magnitude and Phase response of DT-LTI systems.		
Self-study: Derivations of important properties of Fourier Transform, Concept map showing relations among CTFS and DTFS, CTFT and DTFT				
		Total	**45**	

Application areas: Speech signal and model of speech production system, Phonograph, Electrical circuits using resistors, capacitors and operational amplifiers, Hydraulic systems, Mass & Spring suspension system , R-C filters (low pass and high pass – first and second order), Communication systems (e.g. cell phone system), Computer system(e.g. accumulator)

Reference Books:

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Alan Oppenheim, Alan S. Willsky, Syed H. Nawab	Signals and Systems	Prentice Hall	Second/2015
2	Alan V Oppenheim and S Hamid	Signal and System	Person New International Edition	Second/ January 2015
3	Kumar A. Anand	Signals and Systems	Prentice Hall India	Third /2013
4	Simon Haykin	Signals and Systems	Wiley	Second/ 2007
5	Tarun Rawat	Signals and Systems	Oxford Higher Education	First/2010
6	Alberto Leon-Garcia	Probability and Random Processes for Electrical Engineering	Pearson, UKcountry of origin	Second edition, 1993

LAB/TUT CA:	Tutorials includes journal / written record of tutorials. Tutorials will be evaluated out of **25 marks each**. A minimum of 8 tutorials to be performed in a semester based on complete syllabus .

<table>
<tr><th>Course Code</th><th colspan="4">Name of the Course</th></tr>
<tr><td>216U03C405</td><td colspan="4">Electromagnetic Field Theory</td></tr>
<tr><td colspan="5"></td></tr>
<tr><td rowspan="2">Teaching Scheme</td><td>TH</td><td>P</td><td>TUT*</td><td>Total</td></tr>
<tr><td>03</td><td>--</td><td>01</td><td>04</td></tr>
<tr><td>Credits Assigned</td><td>03</td><td>--</td><td>01</td><td>04</td></tr>
<tr><td colspan="5"></td></tr>
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<table>
<tr><td rowspan="4">Evaluation Scheme</td><td colspan="5">Marks</td></tr>
<tr><td rowspan="2">LAB/TUT CA</td><td colspan="2">CA (TH)</td><td rowspan="2">ESE</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>ISE</td></tr>
<tr><td>25</td><td>20</td><td>30</td><td>50</td><td>125</td></tr>
</table>

* Batch wise Tutorial

Course pre-requisites:

- Differentiation Methods
- Vector Algebra, Vector calculus
- Basic laws of electricity.

Course Objectives:

Electromagnetics principles find applications in various allied disciplines of electronics and telecommunication engineering. This course will help the students to learn basic concepts of electromagnetics, related laws, wave equations and applications.

Course Outcomes:

At the end of successful completion of the course the student will be able to

CO1. Apply basic laws of electrostatics in vector form to find electric fields in materials.
CO2. Apply basic laws of magnetostatics in vector form to find magnetic forces in materials.
CO3. Evaluate parameters of dielectric and conducting media by solving wave equations and study polarization of waves.
CO4. Calculate energy transported by means of electromagnetic waves and reflections of plane waves.
CO5. Apply concepts of electromagnetics to real life applications in communications.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	**Electrostatics**		10	CO 1
	1.1	**Electric Field:** Coulomb's law, electric field intensity, charge distributions, electric flux density and Gauss's law		
	1.2	**Electric Potential**: Relation between electric field and potential, electric dipole, electric flux and energy density in electrostatic fields		
	1.3	**Electrostatic field in material:** properties of materials, convection and conduction currents, polarization in dielectrics and dielectric constant.		
	1.4	**Electrostatic boundary:** Boundary conditions, Poisson's and Laplace's equation, resistance and capacitance		
2	**Magnetostatics**		10	CO 2
	2.1	**Magnetic Field:** Biot – Savart's law, Ampere's circuit law, magnetic flux density and magnetic field intensity.		
	2.2	**Energy and Potential:** Magnetic scalar and vector potential, energy in magnetic field, magnetic circuits		
	2.3	**Magnetic Force:** Forces due to magnetic fields, Lorentz force equation.		
	2.4	**Magnetic boundary:** Magnetic boundary conditions, concept of inductance.		
3	**Time Varying Fields and Maxwell's equations**		07	CO 3
	3.1	Faraday's law, transformer and motional EMF, displacement current.		
	3.2	**Maxwell's equations:** Integral and differential form for static and time varying fields and its interpretations, time harmonic fields		
4	**Electromagnetic Wave**		10	CO 4
	4.1	**Wave propagation:** Plane waves in free space, dielectrics and good conductors, Helmholtz equation, properties of plane waves and solution of wave equations.		
	4.2	**Wave parameters:** Intrinsic impedance, propagation constant, skin depth and polarization of waves		
	4.3	**Electromagnetic Power:** Poynting Vector, applications, power flow in plane wave, instantaneous, average and complex pointing vector.		
	4.4	**Wave reflections:** Reflection of plane waves at normal incidence and at oblique incidence, perpendicular and parallel		

		polarization		
5	**Applications**			
	5.1	**Applications of static fields:** Deflection of charged particles, electrostatic generator, magnetic deflection, cyclotron, Hall effect, Ink-jet printer, CRO and electromagnetic pumps.	**08**	**CO5**
	5.2	**Applications of electromagnetic fields:** Transmission lines, waveguides, Antennas, electromagnetic interference and compatibility. biological effects and safety (only construction, working, field distributions and standards)		
		Total	**45**	

Reference Books*

Sr. No.	Name/s of Author/s	Title of Book	Publisher	Edition /Year
1.	William H. Hayt	*Engineering Electromagnetics*	Tata McGraw-Hill, India	7th Edition, 2012
2.	Matthew N. O. Sadiku, S.V.Kulkarni	*Principles of Electromagnetics*	Oxford university press, India	6th Edition, 2015
3.	Bhag Guru and Huseyin Hiziroglu	*Electromagnetic Field Theory Fundamentals*	Cambridge University Press, India	2nd Edition, 2005

Instructions:	Tutorials includes journal / written record of tutorials. Tutorials will be evaluated out of **25 marks each**. A minimum of 8 tutorials to be performed in a semester.

<table>
<tr><th>Course Code</th><th colspan="5">Name of the Course</th></tr>
<tr><td>216U06R406</td><td colspan="5">Open Elective Generic</td></tr>
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<tr><td>03</td><td colspan="2">--</td><td></td><td>03</td></tr>
<tr><td>Credits Assigned</td><td>03</td><td colspan="2">--</td><td></td><td>03</td></tr>
<tr><td colspan="6"></td></tr>
<tr><td rowspan="4">Evaluation Scheme</td><td colspan="5">Marks</td></tr>
<tr><td rowspan="2">LAB/TUT CA</td><td colspan="2">CA (TH)</td><td rowspan="2">ESE</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>ISE</td></tr>
<tr><td></td><td>100</td><td></td><td></td><td>100</td></tr>
</table>

Course Code	Name of the Course			
216U03L401	**Probability, Statistics and Optimization Techniques Laboratory**			
Teaching Scheme (Hrs./Week)	**TH**	**P**	**TUT**	**Total**
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA	**CA (TH)**		**ESE**	**Total**
		IA	**ISE**		
	50	--	--	--	50

Instructions:	
	• LAB CA includes journal / written record of experiments (25 marks) and oral based on experiments performed (25 marks). • Experiments will be evaluated out of **25 marks each**. A minimum of 8-10 will be performed in a semester. • Oral will be taken at the end of the semester related toball experiments for 25 marks. **The total marks for LAB CA will be 50**.

Course Code	Name of the Course			
216U03L402	**Analog Electronics Laboratory**			
Teaching Scheme (Hrs./Week)	**TH**	**P**	**TUT**	**Total**
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	**CA (TH)**		**ESE**	**Total**
		IA	**ISE**		
	50	--	--	--	50

Instructions:	- LAB CA includes journal / written record of experiments (25 marks) and an activity (task) as decided by the incharge faculty (25 marks). - The activity may include Oral/Practical assessments, miniprojects, case study, etc. and will be taken during the semester for 25 marks. **The total marks for LAB CA will be 50**. - Experiments will be evaluated out of **25 marks each**. A minimum of 8-10 will be performed in a semester

Course Code	Name of the Course			
216U03L403	**Communication Systems Laboratory**			
Teaching Scheme (Hrs./Week)	**TH**	**P**	**TUT**	**Total**
	--	**02**	**--**	**02**
Credits Assigned	**--**	**01**	**--**	**01**

Evaluation Scheme	Marks				
	LAB/TUT CA*	**CA (TH)**		**ESE**	**Total**
		IA	**ISE**		
	50	**--**	**--**	**--**	**50**

Instructions:	- LAB CA includes journal / written record of experiments (25 marks) and Evaluation based on experiments performed (25 marks). - Experiments will be evaluated out of **25 marks each**. A minimum of 8-10 experiments will be performed in a semester. - Intermediate oral/evaluation will be conducted in the laboratory for lab CA which will be of total 25 marks. (Lab CA evaluation scheme and detailed rubrics will be declared at the starting of the semester)

Course Code	Name of the Course			
216U03L404	**Advanced Microcontroller Laboratory**			
Teaching Scheme	**TH**	**P**	**TUT**	**Total**
Credits Assigned	-	02	-	02

Evaluation Scheme	Marks				
	LAB/TUT CA	**CA (TH)**		**ESE**	**Total**
		IA	**ISE**		
	50	-	-	-	50

Course pre-requisites:

- Microprocessors and Microcontrollers
- Basic knowledge of C language programming

Course Objectives:

- To introduce the architectural features and application capabilities of MSP430.
- To explore assembly and embedded c programming language on 16 bit and 32 bit platform.
- To understand ARM architecture and its features.
- To design embedded applications using MSP 430 and LPC 2148.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Understand the architecture of MSP 430 microcontroller
CO2. Get Proficiency in embedded C programming language
CO3. Understand the architecture and data processing of ARM
CO4. Apply programming skills to design an application using on chip peripherals of LPC 2148

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	**Introduction to MSP430 microcontroller**		03	CO1
	1.1	Features and Different families and nomenclature of MSP430		
	1.2	Functional block diagram, pin out, memory, memory-Mapped Input and Output, Clock Generator		
	1.3	Overview on CCS		
2	**MSP430 GPIO, Timer and On-chip Peripherals**		10	CO2
	2.1	Digital Input-Output: Non Interruptible I/O and Interruptible I/O: Pin logic diagram Different Control Register, Port register Table		
	2.2	PWM		
	2.3	ADC		
	2.4	Serial Communication		
3	**Introduction to 32 bit microcontrollers**		03	CO3
	3.1	The RISC design philosophy		
	3.2	Architecture (ARM core dataflow model),operating modes, registers		
	3.3	Instruction set for data processing, branching load store, software interrupt, and program status register.		
	3.4	Overview on Keil Micro vision IDE		
4	**ARM LPC 2148 : Overview**		10	CO3, CO4
	4.1	Features, Applications, functional block diagram		
	4.2	GPIO programming		
	4.3	ADC, DAC, timers		
	4.4	UART		
5	**Mini project**		04	CO4
	5.1	Hardware implementation		
	5.2	Software implementation		
	5.3	Debugging and testing		
		Total	30	

Reference Books

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/ Year
1	John Davies	MSP430 Microcontroller Basics	Elsevier	2008
2	Chris Nagy	Embedded Systems Design Using the TI MSP430 Series	Elsevier	1st 2003
3	Andrew Sloss, Dominic Symes, and Chris Wright	ARM System Developer's Guide	Morgan Kaufmann Publishers	1st 2004
4	James A. Langbridge	Professional Embedded Arm Development	Wrox, John Wiley Brand& Sons	2014

Instructions:	● LAB CA includes journal / written record of experiments (25 marks) and oral based on experiments performed (25 marks). ● Experiments will be evaluated out of **25 marks each**. A minimum of 8-10 will be performed in a semester. ● Oral will be taken at the end of the semester related to all experiments for 25 marks. **The total marks for LAB CA will be 50**.